

Option D – Energy and the Environment

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 10       | (a) | (i)  | Use of $\lambda_{\text{max}} = \frac{W}{T}$ (1)<br>$\lambda_{\text{max}} = \frac{2.9 \times 10^{-3}}{288} = 1 \times 10^{-5} \text{ [m]}$ (1)                                                                                                                                                                                                                                                                                            | 1               | 1   |     | 2     | 1     |      |
|          |     | (ii) | Gases allow shorter $\lambda$ solar radiation to pass [where it is absorbed by Earth] (1)<br>Gases absorb [some of the] longer $\lambda$ / infra-red / $10 \mu\text{m}$ radiation [emitted by Earth] (1)<br>Gases emit radiation {in all directions / back to Earth} (1)                                                                                                                                                                 | 3               |     |     | 3     |       |      |
|          | (b) | (i)  | $\text{mass}_{\text{ice}} = 920 \times 3.0 \times 10^{11} = 2.76 \times 10^{14}$ <b>and</b><br>$V_{\text{water}} = \frac{2.76 \times 10^{14}}{1000} = 2.76 \times 10^{11} \text{ (1)}$<br>$2.76 \times 10^{11} \times 20 = 5.52 \times 10^{12} \text{ [m}^3\text{]} \text{ (1)}$<br><br><b>Alternatively for 1<sup>st</sup> marking point:</b><br>$V_{\text{water}} = \frac{3.0 \times 10^{11}}{\frac{1000}{920}} = 2.76 \times 10^{11}$ |                 | 2   |     | 2     | 2     |      |
|          |     | (ii) | Darker areas will absorb more radiation / heat [than lighter areas] (1)<br>Resulting in an increased temperature and rate of ice melting (1)                                                                                                                                                                                                                                                                                             |                 | 2   |     | 2     |       |      |
|          | (c) | (i)  | Hydroelectric relatively constant whereas PV increasing (1)<br>No new hydroelectric sites available or commercially viable / advances in efficiency of PV cells / availability of [domestic] PV cells (1)                                                                                                                                                                                                                                |                 |     | 2   | 2     |       |      |

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks available |             |     |       |       |      |
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|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2         | AO3 | Total | Maths | Prac |
|          |     | (ii) | Use of $P_{\max} = IV$ [at / close to 'knee'] (1)<br>$P_{\max} = 130 \pm 5$ [W] (1)<br>$P_{\text{in}} = IA = 600 \times 1.2 = 720$ [W] (1)<br>$\frac{130 \text{ ecf}}{720} = 0.18$ or 18 [%] (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1               | 1<br>1<br>1 |     | 4     | 3     |      |
|          | (d) | (i)  | Use of $\frac{\Delta Q}{\Delta t} = KA \frac{\Delta \theta}{\Delta x}$ (1)<br>Answer seen 4900 (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1               | 1           |     | 2     | 1     |      |
|          |     | (ii) | Attempt at calculating $\Delta \theta_{\text{for 1 layer}}$ (1)<br>$\Delta \theta_{\text{carpet}} = \frac{1225 \text{ ecf} \times 10 \times 10^{-3}}{0.045 \times 60} = 4.5$ and<br>$\Delta \theta_{\text{underlay}} = 5.8$ and $\Delta \theta_{\text{concrete}} = 3.5$ (1)<br>Add to give 13.8~14 so claim is reasonable (1)<br>Also accept use of $\frac{5\,000}{4} = 1250$ W<br><br><b>Alternative:</b><br>Attempt at calculating $\frac{\Delta x}{KA}$ for 1 layer (1)<br>$R_{\text{T(carpet)}} = \frac{\Delta x}{KA} = \frac{10 \times 10^{-3}}{0.045 \times 60} = 0.0037$ and<br>$R_{\text{T(underlay)}} = 0.0048$ and $R_{\text{T(concrete)}} = 0.0029$ (1)<br>Add to give 0.0114 and $\frac{\Delta Q}{\Delta t} = \frac{\Delta \theta}{\frac{\Delta x}{KA}} = 1228 \approx 1225$ so claim is reasonable (1) |                 |             | 3   | 3     | 3     |      |

| Question |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |          |          |           |           |          |
|----------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|-----------|----------|
|          |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2      | AO3      | Total     | Maths     | Prac     |
|          |  |  | <p><b>Alternative:</b><br/>Attempt at using <math>R</math> values (1)</p> $R_{\text{concrete}} = \frac{0.12}{0.7} = 0.17$ $R_{\text{underlay}} = \frac{0.01}{0.035} = 0.29$ $R_{\text{carpet}} = \frac{0.01}{0.045} = 0.22$ <p>Total <math>R</math> value = 0.68 [K m<sup>2</sup> W<sup>-1</sup>] (1)</p> <p>So <math>R_T = \frac{\frac{\Delta T}{\phi}}{A} \rightarrow \phi = \frac{\Delta T A}{R_T}</math></p> $\phi = \frac{14 \times 60}{0.68} = 1235 \text{ [W] so claim is reasonable (1)}$ |                 |          |          |           |           |          |
|          |  |  | <b>Question 10 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>6</b>        | <b>9</b> | <b>5</b> | <b>20</b> | <b>10</b> | <b>0</b> |